



Exploring how artificial intelligence capabilities impact corporate sustainability performance: Insights from Chinese manufacturing firms

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ABSTRACT

In the context of digital transformation, the driving mechanism of Artificial Intelligence capabilities (AIC) on corporate sustainability performance (CSP) and its organizational boundary requirements have emerged as a key topic in advancing digital sustainability, both in theory and practice. This study is based on dynamic capabilities theory (DCT) and sustainable development theory, and reveals the micro-mechanisms and boundary conditions of AIC driving CSP. By extending the traditional application of AI and dynamic capabilities from competitive advantage to sustainability, this study provides a novel theoretical lens for digital sustainability. This study empirically analyzes data from Chinese manufacturing enterprises through the PLS-SEM method. The findings reveal that AIC significantly enhances CSP through a tiered enabling mechanism, but prescriptive capabilities fail to achieve their intended effects. The type of manufacturing firm constitutes a core boundary condition, and there is significant variability in the sustainable performance of AIC exerted under traditional manufacturing and Industry 4.0 manufacturing firms. The impact of AIC on CSP is significantly weaker in traditional manufacturing firms than in Industry 4.0 manufacturing firms. By adopting a dynamic capabilities perspective, this study examines the role of DCT-based AIC in enhancing CSP, expanding the prevailing technical viewpoint on their interrelationship and providing a compelling justification for integrating AI into digital sustainability frameworks. These findings offer manufacturers actionable insights for harnessing AI technologies to achieve sustainable performance during digital transformation.

1. Introduction

Climate change is reshaping the global economic landscape at an unprecedented pace, pushing industrial enterprises-the primary sources of carbon emissions-to the forefront of sustainable transformation (Voosen, 2025). Global policy trends are rapidly shifting environmental performance from voluntary corporate social responsibility (CSR) to mandatory compliance requirements, such as China's "dual carbon" targets and the European Union's Corporate Sustainability Reporting Directive (CSRD) (Hernández et al., 2024; Bohnsack et al., 2022). However, traditional emission reduction methods such as energy substitution and process optimization have encountered significant technical bottlenecks. The rate of energy efficiency improvement lags far behind the level required to achieve carbon neutrality goals (Chu et al., 2024). This reality highlights the urgency of exploring innovative driving forces. Recently, digital technologies, especially artificial

intelligence (AI), have played an essential role in the transition towards sustainable development (Schwaeke et al., 2025; Bickley et al., 2025; Chen et al., 2025). Therefore, improving industrial sustainability with AI has attracted increasing practical attention (Jiao et al., 2024).

Despite the increasing interest in utilizing AI to improve corporate sustainability performance (CSP), which is defined as a firm's integrated achievements in economic viability, environmental stewardship, and social responsibility (Elkington, 1997; Schaltegger and Wagner, 2011), current studies on the relationship between AI and CSP primarily emphasize qualitative research and lack empirical evidence (Zhou et al., 2025; Schwaeke et al., 2025). Among them, some studies taking the technology perspective elucidate that AI can acquire new knowledge to support managers' rational decision-making, improve resource utilization efficiency (George et al., 2020), and reduce environmental pollution, thus enhancing CSP (Wang et al., 2024; Hussain et al., 2024). However, scholars taking the capability perspective hold that artificial

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intelligence capabilities (AIC), rather than AI technology, perform well in guaranteeing sustainable competitive advantages (Mikalef and Gupta, 2021; Kumar et al., 2025a; Sjödin et al., 2023), which should also be true for CSP. Unlike AI technologies, AIC represents a coherent set of routines integrating tangible resources, such as technology and data, and intangible resources, such as coordination and organizational capabilities, that enable an enterprise to perform specific value-creating activities in a repeatable and reliable manner (Kumar et al., 2025b; Mikalef and Gupta, 2021).

Building on this, and drawing primarily from Sjödin et al.'s (2021, 2023) framework, AIC can be deconstructed into three hierarchical sub-capabilities that represent progressive levels of AI sophistication: perceptive, predictive, and prescriptive capabilities. In general terms, perceptive capabilities focus on sensing and interpreting data in real-time, utilizing algorithms to monitor environments, processes, and systems for immediate insights without forecasting or decision-making elements. This foundational layer enables organizations to gather and analyze current states accurately and efficiently. Predictive capabilities build upon perceptive ones by employing machine learning and statistical models to anticipate future trends, risks, or opportunities based on historical and real-time data patterns. This dimension shifts from mere observation to foresight, allowing firms to prepare for potential scenarios. Finally, prescriptive capabilities extend further by not only predicting outcomes but also recommending optimal actions or strategies through advanced optimization techniques and simulation, providing decision-makers with actionable guidance to achieve desired results. These sub-capabilities, as conceptualized by Sjödin et al., form a coevolutionary framework where they interact and scale together to drive innovation and value creation in various business contexts, such as product-service systems and circular business models (Sjödin et al., 2021, 2023). This general framework lays a foundational structure for examining AIC's relationship with CSP, allowing us to explore how these dimensions contribute to sustainability outcomes. Furthermore, studies have introduced AIC as enabling firms to continuously sense opportunities across markets and technologies, thereby driving performance (Sullivan and Wamba, 2024). However, the role of AIC in enhancing CSP has not been analyzed with empirical evidence. The dynamic capabilities theory (DCT) has been widely used to explain the relationship between digital technology capabilities and corporate sustainable innovation (Heredia et al., 2022), yet it has not been applied to elucidate the AIC-CSP linkage. To fill this gap, this study conceptualizes AIC as dynamic capabilities to investigate how they enhance CSP through rapid information acquisition and opportunity sensing.

Even though AIC can be used to sense opportunities for enhancing CSP, the pathway may not be straightforward and likely depends on its sub-capabilities, an aspect unaddressed in prior research. According to DCT's process view of "sensing-seizing-reconfiguring," opportunities sensed by AIC must be effectively seized and reconfigured to realize CSP gains (Teece et al., 1997). In this vein, dynamic capabilities like AIC not only influence CSP at a higher level but also exhibit differential effects through their sub-capabilities (Shang et al., 2020). Specifically, when linked to sustainability performance, perceptive capabilities can help businesses improve operational efficiency by enabling real-time monitoring of resource use and environmental impacts, as seen in cases like Sany Heavy Industry, which achieved an 8 % efficiency increase and 10 % fuel reduction through such applications (Wenhui et al., 2024). Predictive capabilities, by forecasting trends and mitigating risks, support proactive sustainability strategies, for instance, allowing firms like CATL to reduce production time by 86.5 % and energy consumption by 79.6 % (Wang et al., 2022). Prescriptive capabilities further enhance CSP by offering optimized recommendations for process adjustments that balance economic, environmental, and social goals, exemplified by Haier's process optimizations that boosted overall resource efficiency (Steiber and Alvarez, 2024). Collectively, these sub-capabilities underscore AIC's potential to elevate CSP in manufacturing firms. However, existing studies have not empirically examined the layered effects of AIC and its

perceptive, predictive, and prescriptive dimensions on CSP.

Additionally, this study also examines potential boundary conditions that could either enhance or diminish the impact of AIC on CSP. The manufacturing industry plays a crucial role in the adjustment of China's economic structure (Pan and Zhao, 2019; Shen et al., 2020), which serves as the empirical context for our research. Traditional manufacturing firms encounter even greater difficulties and uncertainties as a result of the industrial sustainable transformation process (Liu et al., 2021, 2023; Yu and Chen, 2025). For manufacturing firms to achieve sustainable development, it is imperative that they reinforce their AIC, foster innovation, and improve their intrinsic development. Xia and Lou (2018) proposed that traditional manufacturing firms rely on linear value chains, highly standardized processes, and locked-in strategic paths, while Industry 4.0 manufacturing firms are characterized by open value networks, agile decision-making structures, and dynamic capabilities investments. AIC may pose a disruptive threat to traditional manufacturing firms, but it may be a catalyst for innovation for Industry 4.0 manufacturing firms. Unfortunately, the industry kinds of organizations-such as traditional manufacturing firms and Industry 4.0 manufacturing firms-have rarely been regarded as significant variables in the literature that currently exists. Therefore, it is necessary to directly explore the influence of manufacturing firm type on relevant variables, to enhance our differentiated understanding of the influence mechanism of AIC on CSP. In this study, we will further explore whether there is heterogeneity in the effect of AIC on CSP among different types of enterprises (traditional manufacturing firms and Industry 4.0 manufacturing firms).

There thus exists a gap in the research concerning: (1) The first empirical test of the overall AIC and its perceptive, predictive, and prescriptive capabilities on the stratified influence mechanism of CSP; (2) Revealing the boundary role of enterprise types in the path of AIC influencing CSP. To reach the research goal, the following research questions:

Research Question 1. How does AIC affect CSP? What is the specific impact of the three sub-capabilities of AIC on CSP?

Research Question 2. Is there significant heterogeneity in the impact of AIC on CSP across different types of industrial firms?

Based on dynamic capabilities theory and sustainable development theory, this study constructs the "AIC→CSP" analytical framework, and introduces the type of enterprise as a boundary condition. To address the above research question, this study uses data from a questionnaire survey of 292 individuals from 120 Chinese manufacturing firms to develop a partial least squares structural equation model (PLS-SEM).

The rest of the sections of this study are organized as follows. In Section 2, a theoretical review and hypothesis formulation are conducted. In Section 3, the research design, methodology, and process are highlighted. In Section 4, the results of the empirical analysis are presented. In Section 5, the findings are discussed in detail, focusing on analyzing the theoretical and practical results of the study. Section 6 summarizes the research contributions, practical implications, and limitations.

2. Theory and hypothesis development

2.1. Dynamic capabilities theory

Dynamic capabilities theory (DCT) emerged as a lens to explain how firms achieve competitive advantage under conditions of uncertainty, by enabling them to integrate, build, and reconfigure internal and external competences to address rapidly changing environments (Teece et al., 1997; Eisenhardt and Martin, 2000). Teece et al. (1997) proposed DCT for fast aligning with the changing external business environment, and dynamic capabilities can be split into three categories: (1) Sensing opportunities, (2) seizing chances, and (3) reconfiguring possibilities. Specifically, sensing opportunities involves scanning activities; seizing opportunities involves agile investment in commercialization activities;

reconfiguring opportunities involves breaking the unfavorable path dependencies that hinder sustainable development (Bashir et al., 2022). For instance, Huawei corporation has identified a new opportunity in low-light photography for mobile phones. To effectively take advantage of this opportunity, Huawei reallocates both internal and external resources. As a result, Huawei launches the P series, which represents a strategic response to this opportunity. This illustrates the process perspective of DCT, demonstrating how firms leverage dynamic capabilities to sustain competitive advantage amid uncertainty.

Research on digital capabilities and innovation has recently made extensive use of DCT (Sullivan and Wamba, 2024; Heredia et al., 2022). In particular, Sullivan and Wamba (2024) investigated how three AI-powered skills may respond to market shifts and grasp opportunities, leading to innovation and improved business performance. Heredia et al. (2022) proposed a model to explain the effect of digital capabilities on firm performance through the PLS-SEM method in the “new normal” context from a firm-level perspective. Previous studies have demonstrated that DCT provides a coherent framework for analyzing the relationship between AIC and CSP. Consequently, we explain how AIC influences CSP by using the process perspectives of DCT, reinforcing that this study’s contribution is theory-driven by extending DCT from traditional competitive advantage to digital sustainability under uncertain conditions.

According to the DCT process view, AIC can be regarded as a high-level capability of an enterprise, as it facilitates the creation of new knowledge and information while also acquiring existing information through various channels (Krakowski et al., 2023). Regarding this, we are reiterating the findings of Hercheui and Ranjith (2020), who discovered that the spread of AIC in manufacturing has an impact on organizations’ dynamic skills in terms of the firm’s ability. Specifically, the sensing capability within dynamic capabilities enables companies to accurately discern market demand, technological trends, and potential risks, thereby identifying opportunities in advance. The seizing capability in dynamic capabilities continuously tracks changes in the environment and the operational status of the enterprise, and promptly detects deviations. The reconfiguring capability in dynamic capabilities can flexibly adjust resource allocation and organizational structure based on feedback from the first two. The three elements work together to help companies respond quickly to the market, optimize operations, strengthen core competitiveness, and ultimately achieve steady growth in sustainable performance. Similarly, the impact of AIC and its three sub-capabilities on CSP also follows the DCT process. By identifying and predicting wasteful processes and procedures in manufacturing, CSP can be seen as an opportunity to reconfigure and take on environmental and social responsibilities (Wang et al., 2025; Madanaguli et al., 2024). Based on the above analysis, it is reasonable for this study to use DCT to explain the research model, as it provides a robust theoretical justification for linking AIC to CSP in uncertain digital transformation contexts.

2.2. Corporate sustainability performance

Corporate sustainability performance (CSP) is conceptually anchored in the Triple Bottom Line framework, which posits that sustainable business outcomes must concurrently advance economic viability, environmental stewardship, and social responsibility (Elkington, 1997; Schaltegger and Wagner, 2011). In this study, CSP is operationally defined as a multidimensional construct comprising three interdependent pillars: first, economic sustainability reflects enhanced operational efficiency through resource optimization and cost reduction (Kumar et al., 2025b), alongside export market development that fosters long-term profitability (Kumar et al., 2025b); second, environmental sustainability encompasses tangible outcomes such as pollution mitigation, waste reduction, and the facilitation of green innovations enabled by technological advancements (George et al., 2020; Hussain et al., 2024; Wang et al., 2024); third, social sustainability manifests in

strengthened stakeholder trust, supply chain reliability (Olan et al., 2024), thereby addressing societal expectations beyond profit maximization. This conceptualization aligns with contemporary sustainability metrics, including the EU Taxonomy for Sustainable Activities and empirical validations in managerial decision-making contexts (Khaled et al., 2021).

The theoretical linkage between CSP and Artificial Intelligence Capabilities (AIC) is fundamentally rooted in Dynamic Capabilities Theory (DCT), which posits that firms leverage high-order capabilities to reconfigure resources and adapt to turbulent environments (Teece, 2007; Mikalef and Gupta, 2021). Here, AIC functions as a meta-capability that integrates tangible resources (e.g., AI infrastructure, data ecosystems) with intangible assets (e.g., organizational routines, human-machine collaboration skills) to transform sustainability challenges into opportunities (Mikalef and Gupta, 2021). Empirical evidence suggests that distinct sub-capabilities within AIC drive specific CSP dimensions: perceptive capabilities enhance decision-making rationality, optimizing resource allocation across economic and environmental spheres (George et al., 2020); predictive capabilities enable proactive pollution reduction and risk mitigation, directly elevating environmental performance (Wang et al., 2024; Hussain et al., 2024); while prescriptive capabilities—though ineffective in this study’s context—hypothetically could advance social sustainability through ethical AI governance (Mancuso et al., 2025). Notably, the AIC-CSP relationship exhibits contingency effects, as Industry 4.0 manufacturing firms harness AIC’s adaptive potential more effectively for social and environmental innovation compared to traditional counterparts (Schwaeke et al., 2025), underscoring the need for boundary conditions in theoretical models.

Critically, this study acknowledges the paradoxical tensions identified in recent literature, where AI-driven sustainability initiatives may inadvertently generate value destruction alongside value creation (Mancuso et al., 2025). For instance, energy-intensive AI computations could negate environmental gains, while opaque algorithmic decision-making might erode stakeholder trust. Such tensions contextualize the non-linear relationship between AIC and CSP, particularly regarding prescriptive capabilities’ limited efficacy in this empirical setting. By grounding the AIC-CSP nexus in DCT and Triple Bottom Line Theory—while integrating empirical validation from manufacturing contexts (Kumar et al., 2025b) and critical perspectives on AI’s dual impact (Mancuso et al., 2025)—this section establishes a robust theoretical foundation for hypotheses development. The clear operationalization of CSP and its mechanistic linkages to AIC’s hierarchical capabilities enhance internal consistency, ensuring that subsequent analyses align with established sustainability literature and dynamic capability paradigms.

2.3. An overview of AIC and CSP

The theory of corporate sustainability emphasizes balancing social and environmental responsibilities while creating economic value, with the goal of long-term survival. It integrates economic, social, and environmental bottom lines, requiring companies to balance short-term interests with long-term development and achieve a win-win situation for all parties through innovation and responsible management (Hussain et al., 2018). CSP takes a broader view of economic, social, and environmental value and shifts the focus from customers and shareholders to multiple stakeholders (Shang et al., 2020; Schwaeke et al., 2025). Although scholars have a wide range of interests, the field of CSP remains in its infancy. Additionally, some scholars explore the drivers of CSP. For instance, scholars theoretically argue that creating shared value (Jadoon et al., 2021), dynamic capabilities (Alkaraan et al., 2024), stakeholder management (Agrawal et al., 2025), blockchain (Xu et al., 2025), and digitalization (Zhang et al., 2023) are significant drivers of CSP. Recently, some scholars regard AI technology as the driver of CSP (Zhong and Song, 2025; Wang et al., 2024). However, these studies are

qualitative and have not yet been supported by empirical evidence, even though there are empirical studies investigating both AI-driven corporate sustainable development (Shang et al., 2022) and sustainability strategies (Treepongkaruna et al., 2025). Furthermore, scholars taking the capability perspective hold that AIC, rather than AI technology, performs well in guaranteeing sustainable competitive advantages (Mikalef and Gupta, 2021), which should also be true for CSP.

As for AIC, it is defined as the organization’s ability to utilize cognitive computing technology to enhance strategic decision-making in various functional areas (Sullivan and Wamba, 2024). From the perspective of dynamic capabilities, AIC can be divided into perceptive, predictive, and prescriptive capabilities (Sjödin et al., 2023). From the perspective of data application, AIC can be divided into data management, algorithm development, and the democratization of AI applications (Sjödin et al., 2021). From the perspective of customer-enterprise interaction, AIC can be divided into symbol recognition, algorithm development, and data pipeline capabilities (Li et al., 2023). Table 1 shows the concepts and classifications related to AIC in recent years. As the core digital capabilities, AIC enables enterprises to promote sustainable development goals (SDGs) through the effective deployment of digital resources and solutions. The interest in AIC can be attributed to its potential to revolutionize various aspects of organizational management, leading to sustainable development, but existing research is mainly qualitative (Di Vaio et al., 2020; Sjödin et al., 2023). Some quantitative studies on AIC investigate its impact on economic performance. For example, based on the DCT, Sullivan and Wamba (2024) find that AIC can promote firm performance and innovation through the response to market changes. Based on the resource-based theory of the firm, Mikalef and Gupta (2021) develop and validate a survey instrument to measure an organization’s AI capability, demonstrating its positive impact on organizational creativity and performance. This provides theoretical and practical implications for understanding and developing AI capabilities in organizations, highlighting the importance of complementary resources beyond just technology. However, there is no empirical research that explores the impact of AIC on CSP, as well as its inner mechanisms and boundary conditions. In summary, this study explores the relationship between AIC and CSP from the perspective of capability, which enriches the driving factors of CSP. Compared with previous studies, this study broadens the research perspective on the relationship between AI and CSP from technology to capability, and incorporates the context of sustainability.

2.4. The relationship between AIC and CSP

AI is playing an increasingly important role in the strategic development of organizations (Zhao and Lee, 2023). According to DCT, we contend that AIC can sense market opportunities, internal resource optimization opportunities, and revenue growth opportunities by integrating inter-firm, internal, and external information, which is conducive to CSP (Nishant and Kennedy, 2020; Sjödin et al., 2023). For instance, Wang et al. (2025) used PLS-SEM and fsQCA methods to propose that AIC has a significant promotional effect on the sustainable development model of manufacturing firms, and verified the mediating mechanism of strategic agility and the moderating effect of environmental dynamics. Jiao et al. (2024) proposed that AIC enables companies to transition to green operations by participating in green and clean innovations, thereby improving their environmental performance. Kumar et al. (2025b) examined how AI capabilities significantly influence export market development and exploration, enabling companies to improve their economic performance. Olan et al. (2024) pointed out the supporting role of explainable AIC in enterprise supply chain decision support, and that data mining and machine learning methods can help improve the automated decision-making process in supply chains. The above research effectively demonstrates the positive impact of AIC on the economic, environmental, and social performance of companies, providing a foundation and framework for the relationship between AIC

Table 1
Relevant studies on AI capabilities.

Literature	Research focus	Relevant argument	Qualitative	Quantitative
Wang et al. (2025)	Classification	Sensing capabilities; seizing capabilities; transforming capabilities	✓	
Sullivan and Wamba (2024)	Classification	AI-enabled automation capability; AI-enabled analytics capability; AI-enabled relation capability		✓
Madanaguli et al. (2024)	Classification	AI infrastructure and platform capabilities; Process automation and augmentation capabilities; Integrated intelligence capability; AI ecosystem orchestration capabilities	✓	
Sjödin et al. (2023)	Definition; Classification	AI capabilities involve leveraging digital technology, such as sensors, connectivity, and advanced algorithms, to provide new value creating and revenue-generating opportunities with the potential for increased profits, ecosystem collaboration, and industrial sustainability. Perceptive capabilities; Predictive capabilities; Prescriptive capabilities	✓	
Li et al. (2023)	Classification	Symbol recognition, algorithm development, and data pipeline capabilities	✓	
Sjödin et al. (2021)	Classification	Data pipeline capabilities; algorithm development capabilities; AI democratization capabilities customer cocreation capabilities; data-driven delivery operation	✓	
Mikalef and Gupta (2021)	Definition; Classification	AI capability is the ability of a firm to select, orchestrate, and leverage its AI-specific		✓

(continued on next page)

Table 1 (continued)

Literature	Research focus	Relevant argument	Qualitative	Quantitative
Wirtz et al. (2018)	Definition	resources. Tangible Resources; human Resources; intangible Resources AI process automation systems; virtual agents; predictive analytics and data visualization; cognitive robotics and autonomous systems; intelligent digital assistants; cognitive security analytics and threat intelligence; identity analytics; edge analytics; machine vision and sensing	✓	

and CSP in this study.

Drawing on the dimensions of AIC by [Sjödin et al. \(2023\)](#), perceptive capabilities enable manufacturers to unlock and create new corporate value by sensing product operational profiles through sensors, developing insights into customer intelligence. Predictive capabilities allow companies to employ AI to convert from trends in prior data to predict future events and behaviors, increasing enterprise value through improved resource utilization. Prescriptive capabilities mainly assist manufacturers in using AI to gain prescriptive insights for optimizing operations at the system level, identifying the ideal actions to maximize sustainable value. AIC ([Sjödin et al., 2023](#); [Olan et al., 2024](#)) has garnered a lot of interest from academics and consistently enables businesses to enhance sustainable performance through resource allocation optimization, innovation, operational efficiency, and decision-making science. Three sub-capabilities within AIC can enhance CSP: slowing, narrowing, and guiding. The specific details are as follows:

The perceptive capabilities of AIC refer to AI using algorithms to monitor product performance in real time and analyze massive amounts of data, enabling companies to instantly identify and resolve issues and improve operational efficiency. They can help manufacturing firms effectively extend the resource use cycle by prolonging the service life of products and their components, thereby slowing down the consumption of industrial raw materials ([Sjödin et al., 2023](#)). For instance, AI can use information from a wide range of connected sensors inside or close to the machines to sense their surroundings ([Ferreira and Goncalves, 2022](#)). The potential to delve deeper into data is presented by AI perception, which eventually yields insights into production intelligence. In addition, AI enables firms to evaluate vast amounts of data and uncover hidden patterns that would be challenging to uncover through conventional research, increasing operational transparency ([Xia et al., 2023](#)). Specifically, manufacturing firms can use perceptive capabilities to monitor product usage and performance in real time, and analyze historical data and behavioral patterns. Implement preventive maintenance measures before potential problems arise, thereby effectively extending the service life of products and minimizing resource waste. Manufacturing firms can rely on slowing mechanisms built on perceptive capabilities to capture anomalies in resource flows within production systems. They can transform wasteful practices such as material redundancy and energy spillover into economic and environmental performance.

The predictive capabilities refer to AI using historical data and pattern recognition technology to predict future trends and outcomes. Through automated recommendations and decision support, enterprises can identify and avoid potential risks in advance and optimize resource allocation. They enable manufacturing firms to use AI to proactively analyze multidimensional data collected, predict anomalies and normal variations, and achieve resource savings and efficiency improvements. Predictive methods analyze trends in historical data and convert the learning outcomes of AI into predictions of future events and behaviors ([Rousopoulou et al., 2022](#)). This enables AI to infer the probability and trends of future events and provide actionable insights for operational planning, such as determining when to perform maintenance shutdowns and developing route planning ([Cao et al., 2022](#)). Early warning of breakdown and resource waste risks is another important advantage of predictive capabilities, enabling businesses to take preventative measures to achieve productivity and sustainability performance ([Sjödin et al., 2023](#)). Therefore, the application of predictive capabilities allows for greater effectiveness through lowering time and material resource waste, boosting productivity, and saving costs. Manufacturing firms can rely on narrowing mechanisms based on predictive capabilities to help them effectively improve the efficiency of material and energy use. Production parameters are automatically adjusted and more energy-efficient operating strategies are proposed, thereby improving sustainability performance.

The prescriptive capabilities refer to AI providing actionable guidance to businesses based on predictive results, enabling them to remain flexible and efficient amid market changes by formulating strategies and adjusting business processes. They help manufacturers build solutions that maximize sustainable value through millions of simulations to identify ideal actions, thereby further optimizing product operations and usage ([Sjödin et al., 2023](#)). Specifically, companies can use prescriptive capabilities to track the entire process of product use and retirement in real time, accurately analyze customer return patterns, and simplify resource recovery processes to reduce resource waste ([Venkatesh et al., 2024](#)). In addition, manufacturers can use prescriptive capabilities to optimize industrial product design (e.g., increasing product disassembly) and promote material reuse ([Jackson et al., 2024](#)). Ultimately, a set of prescriptive and systematic implementation standards and programs will be formed, thereby fundamentally improving industrial production processes. Manufacturing firms can utilize prescriptive capabilities to guide the implementation of optimal system optimization and operation plans, leveraging massive amounts of data and advanced computing capabilities. Based on the analysis above, this study proposes the following hypotheses:

- Hypothesis 1.** AIC has a positive effect on CSP.
- Hypothesis 1a.** Perceptive capabilities have a positive effect on CSP.
- Hypothesis 1b.** Predictive capabilities have a positive effect on CSP.
- Hypothesis 1c.** Prescriptive capabilities have a positive effect on CSP.

2.5. The role of type of manufacturing enterprise

While many studies have examined the effects of firm ownership ([Liu et al., 2025](#)) and size ([Liu and Wang, 2025](#)), few have investigated the impact of firm industry type. According to this study, it is imperative to have discussions about the several kinds of manufacturing companies that are currently in operation, including both traditional and Industry 4.0 manufacturing firms. The relationship between AIC and CSP may differ between the two types of firms. Industry 4.0 manufacturing firms, aligned with the logics of the fourth industrial revolution, integrate cyber-physical systems, Internet of Things (IoT), artificial intelligence (AI), big data analytics, and cloud computing to enable smart, interconnected production ecosystems ([Frank et al., 2019](#); [Ghobakhloo, 2020](#)). This terminology anchors the discussion in widely accepted global frameworks, contrasting with traditional manufacturing firms

that primarily operate under Industrial 2.0 (mechanization) or 3.0 (automation) paradigms.

Traditional manufacturing, as the “foundation industry” of the national economy, is a core component of the industrial system, covering basic industries such as textiles, machinery, chemicals, metallurgy, and building materials (Zhang et al., 2023). Despite the wave of digitalization and green transformation, traditional manufacturing firms continue to support the stable development of China’s economy with their enormous scale, complete industrial chains, and extensive employment capacity. Typical production characteristics of traditional manufacturing firms include standardized production processes coexisting with labor-intensive operations, high energy consumption and high emissions, long supply chains that rely on physical infrastructure, and slow technological renewal and iteration (Liu et al., 2023). For example, industrial energy consumption accounts for more than 65 % of total energy consumption in China, with traditional manufacturing accounting for a large proportion of this. Most traditional manufacturing enterprises are still in the Industrial 2.0 to 3.0 stages, with technological upgrades limited by a shortage of funds and talent. Through technologies such as 5G, industrial Internet, and big data, some traditional manufacturing firms are moving toward full-chain intelligence, gradually transforming into Industry 4.0 manufacturing firms (Tian et al., 2023a). For instance, Foton Lovol Heavy Industry provides remote equipment diagnostic services through an IoT platform (Wang and Chen, 2021), while Guoshun Group has shifted from equipment manufacturing to environmental protection contracting, with service revenue accounting for 85 % of its total revenue. Compared with traditional manufacturing firms, Industry 4.0 manufacturing firms exhibit typical characteristics such as digitalization, greening and low carbonization, service orientation, and business model innovation (Li et al., 2022a; Dalenogare et al., 2018). Table 2 shows the differences between traditional manufacturing firms and Industry 4.0 manufacturing firms.

From a theoretical perspective, the impact of AIC on CSP varies between traditional and Industry 4.0 manufacturing firms, grounded in dynamic capabilities theory (Teece et al., 1997) and the resource-based view (RBV) (Barney, 1991). Dynamic capabilities emphasize a firm’s ability to integrate, build, and reconfigure internal and external competences to address rapidly changing environments, which is more pronounced in Industry 4.0 firms due to their inherent flexibility and innovation orientation. In contrast, traditional firms often exhibit path dependence and resource rigidity, limiting their ability to leverage AIC effectively for sustainability outcomes. This theoretical lens explains the heterogeneity in three key aspects. **First, differences in technical foundations and data integration capabilities.** Traditional factory equipment rooms suffer from data connectivity barriers, making it difficult for AI systems to obtain real-time, complete production data streams. This limits the depth of AI application in energy consumption optimization and predictive maintenance, as per RBV, where insufficient resource integration hampers competitive advantages (Ghobakhloo, 2020). In contrast, the native data integration capabilities of Industry 4.0 manufacturing firms—enabled by cyber-physical systems—allow AI algorithms to directly drive real-time route optimization and dynamic

resource allocation, thereby generating more direct improvements in sustainable performance (Frank et al., 2019). **Second, differences in organizational management.** The pyramid-shaped organizational structure of traditional manufacturing firms constitutes an implicit barrier to AI implementation. For example, rigid departmental barriers hinder the cross-functional collaboration required for AI, and employee resistance to AI reduces the utilization rate of algorithms, reflecting institutional inertia (Werning and Spinler, 2020). Industry 4.0 manufacturing firms adopt flat, network-based organizations, fostering dynamic capabilities that enable AI to achieve a 37 % increase in marginal efficiency for sustainable performance (Rashid et al., 2025; Kolagar et al., 2024). This agility allows for seamless reconfiguration of resources toward sustainability goals. **Third, differences in the depth of AI technology application.** The deployment of AI in traditional manufacturing firms lags behind actual demand. For instance, Meier et al. (2023) argued that traditional factories are forced to prioritize AI for basic efficiency problems due to long equipment downtime, resulting in diminishing returns on sustainability investments. Industry 4.0 manufacturing firms, however, can directly build data-driven models, incorporating energy consumption monitoring and supply chain carbon footprint tracking into the underlying design of AI systems. This creates a multiplier effect through “AI analysis + collaboration” (Kolagar et al., 2024), amplifying CSP via superior dynamic capabilities. Based on the above theoretical and empirical differences, this study proposes the following hypothesis:

Hypothesis 2. The impact of AIC on CSP differs significantly between traditional manufacturing firms and Industry 4.0 manufacturing firms, such that AIC has a significantly weaker positive effect on CSP in traditional manufacturing firms compared to Industry 4.0 manufacturing firms.

A summary of the five hypotheses in the structural path model that will be examined is provided in Fig. 1.

3. Study design and model evaluation

3.1. Study design

This study aims to test the proposed theoretical model and hypotheses, particularly the hierarchical empowerment mechanism of AIC and its boundary conditions affected by enterprise type. To achieve this research objective, we choose to use partial least squares structural equation modeling (PLS-SEM) as the core data analysis method. This method is selected based on the following key considerations and applicability: (1) This research model is exploratory in nature, aiming to reveal the complex path relationship between AIC and its sub-capabilities and CSP, and to examine the role of enterprise type. PLS-SEM is particularly suitable for such exploratory research, especially when theoretical models are still under development or the structural relationships between variables are not yet fully understood (Sarstedt et al., 2019; Hair et al., 2021). Compared with covariance-based structural equation modeling (CB-SEM), PLS-SEM has more relaxed assumptions regarding data distribution and is more adaptable to the non-normal data distribution that may exist in this study. (2) The final effective sample size obtained in this study is relatively limited. PLS-SEM has significant advantages in handling small to medium sample sizes (Hair et al., 2021). PLS-SEM enables effective estimation of path coefficients in complex models even when the sample size is relatively limited (Soumya et al., 2022). This study has a medium-sized effective sample, comprising 120 manufacturing enterprises aggregated at the corporate level, derived from 292 independent questionnaires. According to Hair et al. (2021), the minimum sample size required for PLS-SEM is typically no less than 10 times the maximum number of paths pointing to any latent variable in the model. In this study, the effective sample size far exceeds this minimum requirement, ensuring the robustness and validity of the model estimates. (3) PLS-SEM can simultaneously handle

Table 2

Typical characteristics of traditional manufacturing firms and Industry 4.0 manufacturing firms.

Dimension	Traditional manufacturing firms	Industry 4.0 manufacturing firms
Application of technology	Mechanization, standardization	IoT, AI, Big Data, Cyber-physical systems
Production model	Mass production, rigid processes	Flexible manufacturing, customized production
Supply chain management	Linear, static	Dynamic collaboration, real-time transparency
Driving factors	Capital, equipment	Human capital, data, and knowledge integration

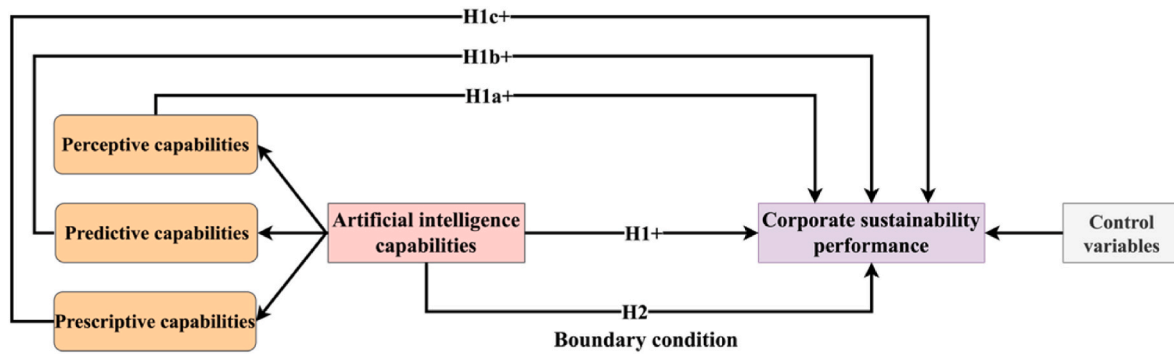


Fig. 1. Hypotheses model.

reflective and formative measurement models. In this study, the core constructs AIC and CSP are both considered reflective constructs, and PLS-SEM can effectively assess their reliability, validity, and interrelationships.

3.2. Data collection

To obtain the sample data needed for this study, we conducted a questionnaire survey from April to September 2024 with 300 senior managers from 120 local Chinese manufacturing companies by adopting the teaching platform of MBA and EMBA at Dalian University of Technology. This platform ensures sample breadth and diversity, while participants' industry expertise allows the data to accurately reflect AIC's role in driving CSP, providing a robust foundation for analysis. Given the focus on AIC in manufacturing enterprises, we selected firms based on China's manufacturing industry standards. The 120 manufacturing enterprises surveyed are widely distributed in the central and eastern regions of China, covering important cities such as Shanghai, Dalian, Hangzhou, Shenzhen, Xi'an, and Beijing. The above cities have strong economic strength, scientific and technological innovation capability, and policy support, and their manufacturing enterprises cover a wide range of areas and face the urgent need for transformation and upgrading, which provides rich practice cases for exploring the innovative path of combining AIC with CSP. The manufacturing sectors of these companies cover a wide range of areas such as automobile manufacturing, electronics manufacturing, textile and garment processing, and steel manufacturing, all of which are closely related to and in line with the direction of development of emerging strategic industries advocated by "Made in China (2025)". During the preparation of the scales and questionnaires, this study specifically invited five senior experts in the field, who thoroughly discussed and meticulously adjusted the reference scales to ensure professionalism. In addition, 50 CEOs in MBA programs were additionally invited to independently review the questionnaire content, and their valuable comments further ensured the high relevance and consistency of the scale content with the design of this study. The questionnaire and scale contents are specified in [Appendix A](#).

To mitigate common methodological bias brought on by the self-assessment method, this study employs a two-stage approach to data investigation. Questionnaires were distributed to multiple senior managers per firm (averaging 2–3 per enterprise) to capture diverse perspectives and enhance data reliability at the firm level (Podsakoff et al., 2003). Responses from the same firm were aggregated by averaging scores across items, transforming individual-level data into firm-level constructs for analysis, as the study focuses on enterprise capabilities and performance. No weighting was applied, as all respondents were equivalently qualified senior managers. Invalid questionnaires (e.g., incomplete, patterned responses) were excluded. The first phase for collecting data on AIC. The second stage is used to collect CSP, and the questionnaire for this stage is distributed only to those who responded to

the completed questionnaires from the first stage. This study considers questionnaires that are filled out incorrectly, omit information, contain substantial portions of the same response, or do not meet the requirements for completion to be invalid. In this study, all questionnaires are coded to make it easier to match the two survey stages. The questionnaires for this study are distributed either face-to-face or by e-mail. In the pre-test survey, 100 questionnaires were distributed in the first stage and 85 were returned, while 85 questionnaires were distributed in the second stage to those who responded in the first stage and 80 were returned, resulting in 75 valid questionnaires. The Cronbach's alpha coefficients of the latent variables were found to be greater than 0.8, and their internal consistency was good. The Bartlett's test of sphericity was significant ($p < 0.001$), and KMO values were higher than 0.8. Therefore, the questionnaire for this study can be formalized. 300 questionnaires were distributed as part of the formal survey's first phase, and 295 valid questionnaires were returned. In the first stage, 295 questionnaires were distributed to participants, and in the second stage, 295 of them were returned. 292 valid surveys were received after invalid ones were removed. The advantages of the partial least squares structural equation modeling (PLS) model in small sample studies are demonstrated by the fact that 292 sample sizes satisfy the model's requirements for sample size.

3.3. Statistical analysis

Table 3 shows the demographic profile of organizations and respondents. Among the group of enterprises participating in this survey, the proportion of state-owned and privately-owned enterprises was significant, with state-owned enterprises occupying 47.82 %, followed by privately-owned enterprises at 25.73 %, while the proportion of foreign-owned enterprises was more limited at 5.49 %. It is further observed that most of the participating enterprises had between 201 and 1000 employees, and this proportion was as high as 86.28 %, which fully reflects the generally moderate size of the participating enterprises in the survey. In addition, these enterprises were mainly concentrated in four major industries, i.e. electronic manufacturing, automobile processing, textile processing, and iron and steel manufacturing, with the electronic manufacturing industry accounting for the highest proportion of 31.13 %, followed by motor vehicle (23.48 %), textile processing (22.73 %) and iron and steel manufacturing (16.91 %), which constitute the major groups of enterprises in this survey. The main group of participants in this survey were from the senior management of the company, with the CEO (Chief Executive Officer) and CSO (Chief Strategy Officer) reaching 27.72 % and 15.82 %, respectively. These senior executives not only have a deep understanding of and control over the daily operations of their companies, but also play a leading role in the practice of strategic transformation. Their participation ensures the professionalism and depth of the survey and is fully in line with the core theme of this study, which focuses on corporate innovation strategy and transformation. In addition, the proportions of traditional

Table 3
Demographic profile of organizations and respondents.

Item	Categories	%	Item	Categories	%
Profile of organizations:					
Nature of company	State owned	47.82	Sector	Motor vehicle	23.48
	Private	25.73		Electrical	31.13
	joint venture	20.96		Textile	22.73
	Foreign owned	5.49		Steel	16.91
Employees	Below 200	7.34	Type of enterprise	Others	5.75
	201–500	48.47		Traditional manufacturing firms	52.35
	501-1000	37.81		Industry 4.0 manufacturing firms	47.65
	Above 1000	6.38			
Profile of respondents:					
Position	Chairman	4.88	Education	PHD	2.38
	CEO	27.72		Masters	34.59
	COO	20.37		Bachelors	58.92
	CFO	14.48	Place of education	College	4.11
	CTO	16.73		Local	90.29
	CSO	15.82		Overseas	9.71
Years of experience	Below 10 years	11.29	Age	Below 30 years	3.19
	10–15 years	46.79		31–40 years	40.93
	15–20 years	34.97		41–50 years	48.92
	Above 20 years	6.95		Above 51 years	6.96
Gender	Male	68.82			
	Female	31.18			

Notes: CEO-Chief Executive Officer; COO-Chief Operating Officer; CFO-Chief Financial Officer; CTO-Chief Technology Officer; CSO-Chief Strategy Officer.

manufacturing firms and Industry 4.0 manufacturing firms account for 52.35 % and 47.65 %, respectively.

In this study, data for all variables were collected from the same questionnaire, and this centralized source of data may lead to a potential risk of homology variance (Gabriel et al., 2019). This study specifically adopts Harman's one-way test for rigorous statistical analysis of the data collected from the questionnaire to ensure the rigor and accuracy of the study for examining potential homology variance. Following Podsakoff et al. (2003), the percentage of a total factor's eigenvalue to the covariance of all variables is regarded as an indicator for assessing homology variance, and above 50 % there is a serious homology variance. As analyzed by SPSS 21, the loading of the first principal component without rotation is 24.49 %, which is below the threshold value of 50 %, indicating that there is no single factor explaining most of the variance in the data collected in this study, and therefore it can be considered that the problem of homology bias in this study is not serious.

Based on our empirical analysis, 3.2 % of the values in the dataset were missing, compared to a median missingness rate (MDN) of 1.9 % and an interquartile range (IQR) between 0.9 % and 4.2 %. Further assessment showed that 86 % of the indicators had missing rates below the 5 % threshold set by the "rule of thumb". Only two indicators had missing rates slightly above 10 % (i.e. 11 % and 13 %), but neither of these exceeded the 15 % explicit exclusion threshold. Given the current lack of established methods for combining multiple PLS-SEM results from different MICE (Multiple imputation by chained equations) estimation datasets, we employed a standard mean imputation strategy to deal with these missing values.

Sample size plays a pivotal role and is an indispensable consideration in the process of model path analysis. According to the requirements of this study, the sample size must meet a specific minimum threshold criterion 30 to ensure the robustness and validity of the analysis, i.e., not less than the value obtained by multiplying the maximum number of structures of the latent variable by 10 (Hair et al., 2021). In addition, the lowest path coefficient in the structural equation model is -0.13 , which as an input parameter gives results that significantly exceed the expected estimates based on the inverse square root method. The above analysis further enhances the reliability and accuracy of the model results.

3.4. Measures

In this study, the scales applied were obtained using two main

acquisition methods. First, direct reference was made to proven and mature scales from existing studies, which cover important dimensions such as CSP. Then, based on the in-depth understanding and analysis of the existing scales in the previous studies, the scales are modified and supplemented in the key areas of AIC to ensure that they are closer to the actual needs of this study and have higher applicability.

AIC has great potential to promote sustainable development by reducing and eliminating traditional inefficiencies at the system level. The construct on AIC was adopted from Sjödin et al. (2023), who based their framework on case studies of multiple observations of complex organizational processes. The study divided AIC into three sub-dimensions treated as reflective latent constructs in the PLS-SEM model: perceptive capabilities, predictive capabilities, and prescriptive capabilities. Perceptive capabilities are the continuous monitoring of device performance to identify hidden patterns and anomalies. Predictive capabilities are the abilities to anticipate and respond to environmental changes from multidimensional data to enable efficient resource utilization. Prescriptive capabilities involve running simulations to determine the ideal actions to maximize sustainable value. To operationalize these, the scale items were adapted and integrated with manufacturers' production processes during initial testing to reflect comprehensive firm-level digital practices. For example, a perceptive capability item might be: "Continuously monitoring equipment performance" (coded on a five-point Likert scale). Responses to these items were transformed by averaging scores across multiple respondents per firm to form the latent variable scores for each sub-dimension.

CSP was measured using three dimensions, namely economic performance, environmental performance, and social performance, mainly referring to the categorization of CSP by Shang et al. (2020). Economic performance emphasizes the profitability of a company by working with its stakeholders. Environmental performance focuses on eliminating adverse environmental impacts from production processes and building an environmentally friendly corporate image. Social performance is satisfying the interests of all parties by improving the health of employees or the community. The scale contains a total of 18 observed variables related to corporate profit, streamlining processes, handling waste, and improving employee health. These dimensions were modeled as reflective latent constructs in PLS-SEM, with items such as: "Our firm has improved profitability through stakeholder collaboration" for economic performance, or "Reduced environmental impacts of production processes or eliminated environmentally damaging processes" for

environmental performance. Variable construction involves aggregating and coding responses in a manner consistent with AIC processing.

For the consistency of the model measurements, this study uses five-point scales to measure the variables, including AIC and CSP. The use of a five-point scale for questionnaire surveys on measures involving complex concepts such as AIC has advantages and applicability. This scale not only reduces the cognitive load of the respondents, but also provides a more comprehensive picture of the differences in respondents' views and attitudes, providing strong data support for the study. Specifically, all items were coded on a Likert scale ranging from 1 (Strongly Disagree) to 5 (Strongly Agree). To transform questionnaire responses into model variables, a two-stage data collection process was employed to mitigate common method bias: the first stage collected data on AIC, and the second stage collected CSP data only from respondents who completed the first stage. Per firm, responses from multiple senior managers (averaging 2–3) were aggregated by averaging item scores without weighting, as all were equivalently qualified. Invalid responses (e.g., incomplete or patterned) were excluded. This aggregation converted individual-level data to firm-level constructs, ensuring the PLS-SEM analysis reflects enterprise-level capabilities and performance. The results of the model evaluation are presented in the following sections along with a more thorough description of the measurement and structural model.

3.5. Control variables

In terms of the consideration of control variables, this study pays particular attention to the selection of several key variables to ensure the rigor of the study. At the firm level, given the possible limitations of small and medium-sized enterprises (SMEs) in addressing the challenges of sustainable development, this study selects firm size and firm age as control variables, and logs the firm's assets and time in business, respectively. At the industry level, this study employs the assessment of senior executives to segment industry concentration into low, medium, and high levels to assess competitive behavior and avoid the potential impacts of concentration (Dupire and M'Zali, 2018) and to ensure the comprehensiveness and accuracy of the study.

3.6. Measures model

The measurement model in this study consists of two components, of which AIC and CSP belong to the reflective-reflective structure. This study will measure the different scale structures as follows.

In this study, two reflective variables are measured, and the loading values of their 27 observed variables are higher than 0.70, with loadings in the range of 0.74 (EP01) and 0.96 (NP08) (see Table B1 in Appendix B). In addition, the internal consistency reliability of the variables was also assessed, with Cronbach's alpha above 0.7 for all variables (see Table B2). Then, the average explained variance (AVE) of all reflective structural variables is above 0.5, indicating good convergent validity (see Table B2). The construct intercorrelations in the model do not

surpass the square root of the AVE, as Table 4 demonstrates. As shown in Table 4, the square root of AVE along the diagonal of the correlation matrix exceeds the off-diagonal correlation values, indicating satisfactory discriminant validity. Finally, this study analyzes the discriminant validity of all reflective structural variables. According to Table B3, the bootstrapped HTMT values (i.e., the upper bounds of the 90 % confidence interval) deviate considerably from 1 and all hetero-trait–monotrait (HTMT) values fall below 0.9.

For the structural model measure, this study shows that the model produces a convergence effect after 15 iterations and the value of VIF is between 1.19 and 1.92, which is significantly lower than the commonly set threshold of 3, indicating that there is no covariance problem. In addition, with reference to Hair et al. (2021), 10,000 samples are used in this study to estimate the standard errors and confidence intervals of the model.

4. Analyses and results

4.1. Assessment of structural model

The underlying hypotheses are tested using the full sample, without distinguishing between traditional and Industry 4.0 manufacturing firms. The path coefficients and hypothesis testing results in the structural model are shown in Table 5 and Fig. 2. The empirical results show that all three model paths are significant at the p-value less than 0.01 level ($p < 0.01$), except for the relationship between prescriptive capabilities and CSP (H1c), which is not significant. Thus, three hypotheses in this study are supported, which are H1, H1a, and H1b.

Hypothesis 1 proposes a positive relationship between AIC and CSP. The results in Table 5 show that AIC has a positive effect on CSP ($\beta = 0.49$; $p < 0.001$). Indeed, the AIC of firms may positively influence CSP by optimizing operational processes and enhancing efficient resource allocation, providing explanatory power for these results. In hypotheses 1a and 1b, the coefficients β of the impact of perceptive capabilities and predictive capabilities on CSP reach 0.47 ($p < 0.01$) and 0.43 ($p < 0.01$), respectively, which are significant at p-values below 0.01. In hypothesis 1c, the coefficient of the impact of the firm's prescriptive capabilities on CSP is 0.28, with a p-value higher than 0.05, which does not prove the significance of the relationship. Possible explanations are that prescriptive capabilities are higher-order capabilities based on massive data analysis and decision simulation, and that their formation and development are constrained by the operational processes of industrial manufacturing. Given that manufacturing firms do not yet have well-developed prescriptive capabilities, they may not be able to realize the role of prescriptive capabilities on CSP in the short term.

4.2. Main effects analysis under different types of manufacturing firm

In hypothesis 2, type of manufacturing firm is used as an organizational boundary, and manufacturing firms are classified into traditional manufacturing firms and Industry 4.0 manufacturing firms to analyze the impact of AIC on CSP separately. Table 6 shows the results of the impact of AIC on CSP under the traditional manufacturing samples and Industry 4.0 manufacturing samples. In the traditional manufacturing samples, the positive effect of AIC on CSP is significant, but the coefficient is only 0.29*, which is lower than the full sample result ($\beta = 0.49$; $p < 0.001$), as shown in Table 6. In terms of the three dimensions of AIC, perceptive capabilities have a relatively weak effect on CSP ($\beta = 0.20$; $p < 0.05$), predictive capabilities play a lower role than the full sample in CSP ($\beta = 0.18$; $p < 0.05$), and prescriptive capabilities have a coefficient of -0.13 , which is not only non-significant but also the exact opposite of the full sample result.

Distinguishing from h traditional manufacturing firms, the positive effect of AIC in the context of Industry 4.0 manufacturing firms is significant and stronger, with a coefficient higher than the full-sample result at 0.58 and significant at $p < 0.001$. The coefficients of the

Table 4

Intercorrelations of the latent variables for the first-order constructs and square root of average variance extracted.

	PC1	PC2	PC3	EP	NP	SP
PC1	0.77					
PC2	0.62**	0.78				
PC3	0.51*	0.28***	0.79			
EP	0.63*	0.33*	0.12*	0.79		
NP	0.39**	0.26**	0.19*	0.32*	0.78	
SP	0.46*	0.32	0.33**	0.38***	0.28**	0.76

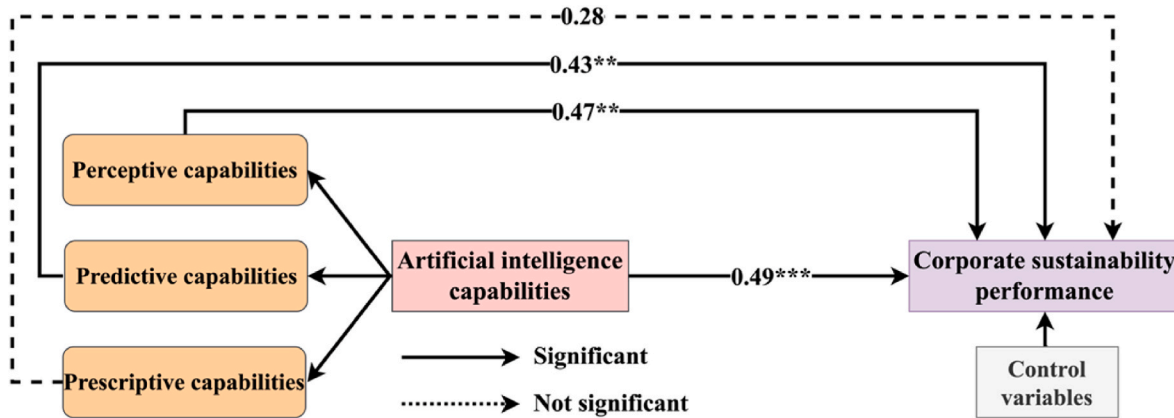
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$ (two-tailed); $N = 292$. PC1: Perceptive capabilities. PC2: Predictive capabilities. PC3: Prescriptive capabilities. EP: Economic performance. NP: Environmental performance. SP: Social performance.

Table 5

Bootstrapped paths of the final model.

Path	Original estimate	Bootstrap mean	Bootstrap SD	T	2.5 % CI	97.5 % CI	Support
H1: AIC→CSP	0.49	0.52	0.03	5.38***	0.38	0.61	✓
H1a: PC1→CSP	0.47	0.50	0.05	2.97**	0.37	0.59	✓
H1b: PC2→CSP	0.43	0.45	0.07	2.88**	0.36	0.57	✓
H1c: PC3→CSP	0.28	0.31	0.09	1.86	0.22	0.32	×

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$ (two-tailed). AIC: Artificial intelligence capabilities. PC1: Perceptive capabilities. PC2: Predictive capabilities. PC3: Prescriptive capabilities. CSP: Corporate sustainability performance. ✓: Hypothesis is supported. ×: Hypothesis is not supported.

**Fig. 2.** Path coefficients for structural modeling.**Table 6**

Path coefficients for stratified analysis of manufacturing firms.

Path	Traditional manufacturing firms		Industry 4.0 manufacturing firms		Absolute value of difference in coefficients
	Original estimate	T	Original estimate	T	
H1: AIC→CSP	0.29	1.99*	0.58	7.38***	0.29
H1a: PC1→CSP	0.20	2.13*	0.49	3.11**	0.29
H1b: PC2→CSP	0.18	2.11*	0.46	6.21***	0.28
H1c: PC3→CSP	-0.13	1.62	0.30	2.04*	0.43

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$ (two-tailed). AIC: Artificial intelligence capabilities. PC1: Perceptive capabilities. PC2: Predictive capabilities. PC3: Prescriptive capabilities. CSP: Corporate sustainability performance.

impact of perceptive and predictive capabilities on CSP increased to 0.49 ($p < 0.01$) and 0.46 ($p < 0.001$), respectively. Unexpectedly, the coefficient of prescriptive capabilities on CSP reaches 0.30 and is significant at $p < 0.05$.

4.3. Effects of control variables

Among the three control variables, firm size is significantly and positively correlated with AIC ($\beta = 0.31$; $p < 0.01$), which suggests that firm scale not only empowers AIC through capital and technology reserves, but also amplifies technology dividends through ecological location advantages. This supports the development of the resource base view (Barney, 1991) in the digital sphere. Firm age shows a positive effect on CSP ($\beta = 0.17$; $p < 0.05$). Industry concentration is significantly positively correlated with CSP ($\beta = 0.19$; $p < 0.05$), which reveals a divergence of sustainable transformation paths in monopolistic markets. In highly concentrated industries, head players rely on market dominance to internalize AIC and achieve sustainable premiums through technology exclusivity. Small and medium-sized enterprises (SMEs), on the other hand, have fallen into a “sustainable depression” due to limited ecological collaboration. In AI-driven sustainable transformation, industry structure reconfigures the road to value capture through competitive norms, while enterprise scale establishes the groundwork for technology adoption.

5. Discussion

5.1. Effect of AIC on CSP

The empirical findings indicate that AIC has a significant positive driving effect on CSP overall (H1 is established), which confirms the theoretical expectation of treating AIC as dynamic capabilities. That is, AIC can empower sustainable development of enterprises through information integration and opportunity perception, which is consistent with the views of Wang et al. (2025), Jiao et al. (2024), and Kumar et al. (2025b). However, an in-depth analysis of the hierarchical effects of AIC sub-capabilities reveals a more complex internal mechanism: perceptive capabilities and predictive capabilities, as the foundation of technology empowerment, have been proven to effectively and directly improve CSP (H1a and H1b are valid). This result is consistent with the classic logic of “instrumental rationality dominates short-term gains” in technology empowerment theory (Wu and Fan, 2025; Phillips, 2019). Specifically, solving specific problems through hardware performance upgrades and data training efficiency can quickly translate into observable and sustainable performance gains (Ghobakhloo et al., 2023).

Contrary to prior studies (Sjödin et al., 2023), our data shows the relationship between prescriptive capabilities and CSP is not significant. This discrepancy may stem from the phased disconnection and

divergence of paths between AI technology and AIC accumulation in the digital transformation process of manufacturing firms. Specifically, compared with prescriptive capabilities, perceptive capabilities eliminate the “information black box” through real-time data feedback, directly reducing the decision-making lag costs of manufacturing firms. With the continuous accumulation of data collected by manufacturing firms, sustainable performance will be driven in a stable manner. Predictive capabilities are based on inferences about future trends based on perceptual data, forming a positive cycle with sustainable performance. However, prescriptive capabilities, as high-level capabilities, require breaking through organizational inertia and interest barriers (Li et al., 2022b, 2024). In addition, the benefits of prescriptive capabilities are highly indirect and long-term, and are easily disrupted by external variables. For instance, an automotive manufacturer deployed AI-driven process optimization algorithms, but the approval system and departmental barriers resulted in a disconnect between the algorithmic models and production requirements. This echoes Teece et al. (1997) assertion that the value of high-level dynamic capabilities may be constrained by organizational foundations. This may hinder the impact of higher-order prescriptive capabilities on sustainable performance. This discovery offers a dynamic, multi-layered viewpoint for studying AIC. That is, the short-term focus should be on the instrumental value of perceptive and predictive capabilities (e.g., equipment monitoring to reduce consumption), while the long-term goal should be to use organizational reengineering to unlock the systemic potential of prescriptive capabilities.

5.2. AIC value transformation mechanisms for different types of manufacturing firms

This study introduces enterprise type as a boundary condition, profoundly revealing the impact of different types of manufacturing firms on AIC value transformation, and providing empirical evidence for understanding the context dependence of AIC empowering CSP. Empirical results show that in traditional manufacturing firms, the promotional effect of AIC on CSP is significantly weakened, and its hierarchical empowerment effect exhibits characteristics of “marginal diminishing returns.” Although perceptive and predictive capabilities remain effective, their effects are far less than those of Industry 4.0 manufacturing firms, while prescriptive capabilities are insignificant and even exhibit negative values in the opposite direction. This deeply reflects the inherent constraints of traditional manufacturing models in terms of management processes (Tian et al., 2023b), departmental barriers (Werning and Spinler, 2020), and path dependence (Brekke et al., 2024). These constraints can be further understood through the lens of absorptive capacity theory (Cohen and Levinthal, 1990; Zahra and George, 2002), where traditional manufacturing firms’ limited prior experience with digital technologies hampers their ability to recognize, assimilate, and apply AIC innovations, thus restricting its transformation into sustainable outcomes. For example, an automotive manufacturer invested 70 % of its AI budget in production line optimization, but the multi-level approval system resulted in a technology iteration cycle of up to 18 months, ultimately rendering the effects of low-carbon process design insignificant. That is, AIC has limitations as dynamic capabilities in the context of traditional manufacturing firms.

Conversely, Industry 4.0 manufacturing firms, with their open value networks (Awan et al., 2021), flat organizations (Iram et al., 2024), data-driven and agile decision-making organizational characteristics (Khin and Kee, 2022), provide a natural “soil” for the release of AIC value (especially prescriptive capabilities). However, this superiority extends beyond mere organizational openness or ecosystem participation. Industry 4.0 firms often possess pre-existing capabilities to handle digital technologies, which significantly enhance their absorptive capacity for AI (Cohen and Levinthal, 1990). This prior knowledge base allows them to more effectively integrate AIC into core processes, amplifying its impact on sustainability by facilitating the assimilation of

external AI innovations and their internal reconfiguration for CSP gains (Zahra and George, 2002). In this context, AIC not only plays a significant role in promoting CSP, but also enables each sub-capability to effectively fulfill its intended function. For instance, Industry 4.0 manufacturing firms can deeply embed sustainable goals such as energy consumption monitoring and carbon footprint tracking into the underlying design of AI systems, and achieve real-time routing optimization and dynamic resource allocation through cross-system data flows, ultimately improving sustainable performance (Wenhui et al., 2024; Wang et al., 2022; Steiber and Alvarez, 2024). Therefore, the fundamental difference between the two types of enterprises lies not only in the depth of technology application, but also in whether the organizational infrastructure can provide the necessary “adaptability” for AIC (especially prescriptive capabilities). This adaptability is theoretically nuanced by absorptive capacity, where Industry 4.0 firms’ enhanced ability to absorb AI knowledge acts as a multiplier for dynamic capabilities, distinguishing them from traditional counterparts. This enables it to transform from a passive response tool into a strategic lever that proactively drives systematic and sustainable change. This finding greatly enriches our understanding of the boundary conditions for the value conversion of digital capabilities, highlighting the central role of organizational structure and management models in unlocking the full potential of AIC.

6. Conclusion

6.1. Theoretical implications

- (1) This study empirically tests the hierarchical influence mechanism of AIC and its sub-capabilities on CSP, significantly expanding the existing technology-dominated perspective on the relationship between AI and sustainability. It takes the lead in deconstructing and verifying AIC under the dynamic capabilities theory (DCT) framework, revealing its complexity as high-order dynamic capabilities driving CSP. Previous studies have mostly focused on technical functionality, broadly assuming that AI promotes sustainability by improving decision-making rationality and resource efficiency, but lacking empirical analysis of the composition of AIC and its differentiated pathways. Based on DCT, this study not only confirms that AIC has a positive impact on CSP overall, but more importantly, through hierarchical analysis, it reveals that “perceptive capabilities” and “predictive capabilities” as basic technical enabling layers can directly and effectively improve CSP, while the role of “prescriptive capabilities” as a high-level system optimization layer has not met expectations. This empirical finding reveals that AIC-enabled CSP is not a homogeneous “black box” process, but rather follows the logic of “instrumental rationality first, value rationality second.” Of these, instrumental rationality emphasizes the efficiency of behavioral means, with the goal of maximizing efficiency. In contrast, value rationality emphasizes the social value of behavior, that is, the long-term value of behavior to the public interest and moral norms (Wu and Fan, 2025; Phillips, 2019). This finding not only challenges Sjödin et al. (2023)’s assumption that AIC sub-capabilities are equally effective contributing factors, but more importantly, it bridges a key explanatory gap in dynamic capabilities theory in organizational contexts: the potential value of higher-order dynamic capabilities (such as prescriptive capabilities) may be offset by organizational infrastructure rigidity. Therefore, this study uses hierarchical empirical testing to not only deepen the research perspective on AI and sustainability from “technology application” to “capability building,” but also explores the boundaries and conditions of AIC at different levels. This provides a more organizationally relevant, hierarchical theoretical foundation for dynamic capabilities

theory in explaining the effectiveness release mechanism of digital capabilities (especially AIC).

- (2) This study reveals that the type of enterprise is a boundary condition that significantly affects the efficiency of the transformation from AIC to CSP, providing important contextual deepening and boundary expansion for the application of dynamic capabilities theory (DCT) in the digital context. Empirical results show that the value realization of AIC does not exist in isolation, but is deeply embedded in specific organizational structures and operating models. Traditional manufacturing firms are constrained by their linear value chains, rigid processes, departmental barriers, and historical path dependence. Even if they invest resources in developing AIC, its effectiveness is greatly weakened by organizational “organizational inertia,” making it difficult for AIC to effectively transform into CSP. This echoes and deepens the classic argument in organizational theory that “organizational inertia” hinders change (Hannan and Freeman, 1984; Lyu et al., 2016). On the contrary, Industry 4.0 manufacturing firms rely on their open value networks, flat organizations, agile decision-making, and data-driven characteristics to provide a natural “fertile ground” for the effective operation of AIC (especially high-level prescriptive capabilities), thereby significantly amplifying the positive impact of AIC on CSP. This finding not only confirms the key role of “organizational adaptability” in the realization of capability value in dynamic capabilities theory (Teece et al., 2016), but more importantly, it clearly identifies and empirically validates “enterprise type” as a core contextual variable and theoretical boundary for understanding differences in the effectiveness of digital capabilities (such as AIC). This fills a theoretical gap in previous research, which has not sufficiently addressed the heterogeneity of organizational infrastructure and its far-reaching implications when exploring the relationship between digital capabilities and sustainable performance. It provides a key theoretical lens and organizational-level explanatory mechanism for future research to understand “why the same digital capabilities produce vastly different results in different companies.”

6.2. Practical and managerial implications

This study finds that AIC is key enablers for industry to move towards sustainability, and calls for more managers of manufacturing companies engaged in digital transformation to focus on and capitalize on that potential. Primarily, companies realizing sustainable performance need to rely on AIC that facilitates sustainability opportunity perception and event prediction throughout the accelerated commercialization process. Although managers usually view sustainability as a transformational direction and give importance to AI technologies, this study shows that AIC has a positive impact on the sustainable performance of industrial firms. By emphasizing strategic innovation and change in the age of AI, AIC helps generate sustained performance. Thus, managers should view the development of AIC as an opportunity to improve CSP. Traditional manufacturing enterprises should focus on the modular embedding of perceptive and predictive capabilities, and direct AI algorithms into local optimization scenarios such as energy consumption monitoring and dynamic scheduling of supply chains. Industry 4.0 manufacturing firms need to strengthen the eco-level synergy of their prescriptive capabilities, but they need to be wary of the hidden environmental costs of arithmetic expansion. AI investment must be closely linked to organizational transformation for both kinds of businesses. By separating their extremely complicated business segments, traditional companies create mechanisms for agile development.

Then, managers need to recognize that AIC is a key enabling factor in driving manufacturing companies to improve CSP, but its value is not uniformly released and is highly dependent on the type of enterprise. First, managers of manufacturing enterprises should recognize that the

improvement of AIC requires a layered focus strategy. Perceptive capabilities and predictive capabilities, as the technical foundation layer, can directly and quickly bring about significant and sustainable performance improvements by improving operational efficiency, and should be the focus of current investment. Companies can prioritize the application of AI in scenarios dominated by “instrumental rationality,” such as energy consumption monitoring, dynamic supply chain scheduling, and predictive maintenance, to obtain short-term visible benefits. Next, managers need to accept that prescriptive capabilities won’t work on their own in the short term. Since these are high-level capabilities, they require a lot of organizational inertia and cross-departmental coordination, so their value is often indirect and long-term. Therefore, companies should not expect to see immediate improvements in sustainable performance simply by deploying complex AI decision-making systems. Instead, they should view this as a long-term capacity-building goal and recognize that realizing its full potential will require deep organizational process reengineering and management model transformation. Finally, and most importantly, companies must deeply recognize that their own type is the core boundary condition for AIC value transformation. Traditional manufacturing companies are constrained by data silos, rigid processes, and organizational structures, resulting in significantly lower efficiency in the transformation of their AIC to CSP compared to Industry 4.0 manufacturing firms. In summary, manufacturing enterprises need to formulate differentiated AIC development paths and organizational support strategies based on their own characteristics to maximize the enabling value of AI technology for sustainable transformation.

6.3. Limitations

This study has several limitations. First, the industry coverage and geographical distribution of the sample may limit the generalizability of the findings. The study data are mainly from Chinese manufacturing firms, especially focusing on the dichotomous comparison between traditional and Industry 4.0 manufacturing firms. While this enables companies to analyze the transformation process in depth in the Chinese context, it may limit the universality of the conclusions in other regions of the world, other industries (such as the service industry), or different types of companies (such as multinational companies). In addition, companies may be at different stages of digital transformation in reality, and a simple dichotomy may not fully capture the complex mix of forms in between. Future research needs to include more granular indicators of corporate digital maturity. Furthermore, the research model does not sufficiently analyze dynamics and time effects. This study is mainly based on cross-sectional data and uses static analysis methods. However, the construction and improvement of AIC and its impact on CSP are likely to be a dynamic process that may even involve lag effects. The lack of tracking and analysis of the time lag between capacity development and performance impact is a shortcoming of this model in explaining long-term dynamic relationships. Finally, the assessment of digital capabilities is relatively narrow. The study focuses on the direct impact of AIC on CSP and the boundaries of their enterprise types. However, the sustainable transformation of enterprises in reality may be driven by the synergistic combination of multiple digital capabilities (such as big data analysis capabilities, IoT capabilities, and digital platform capabilities), which may be complementary or substitutable. This study has not yet explored the interaction between AIC and other key digital capabilities and their comprehensive impact on CSP, which limits the depth of understanding of the overall digital empowerment of enterprises.

CRediT authorship contribution statement

Chen Renfei: Project administration, Investigation, Formal analysis, Conceptualization. **Li Zhongwen:** Methodology, Formal analysis, Data curation. **Yang Guangfei:** Supervision, Resources. **Li Wenli:** Visualization, Data curation. **Guo Zitong:** Supervision, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Measurement scales

Constructs
Artificial intelligence capabilities (Sjödin et al., 2023)
1. Perceptive capabilities (PC1)
PC101: Continuously monitoring equipment performance
PC102: Recognizing real-time usage patterns & anomalies
PC103: Augmented transparency & problem solving
2. Predictive capabilities (PC2)
PC201: Proactively anticipate and react to changes
PC202: Forecasting future outcomes
PC203: Automated suggestions & decision making
3. Predictive capabilities (PC3)
PC301: Real-time problem solving
PC302: Improving proactive risk management
PC303: Augmenting decision making with AI prescriptions
Corporate sustainability performance (Shang et al., 2020)
1. Economic performance (EP)
EP01: Worked with government officials to protect the company's interests
EP02: Reduced costs of inputs for same level of outputs
EP03: Sold waste products for revenue
EP04: Created spin-off technologies that could be profitably applied to other areas of the business
2. Environmental performance (NP)
NP01: Chose inputs from sources that are remediated or replenished
NP02: Reduced environmental impacts of production processes or eliminated environmentally damaging processes
NP03: Reduced operations in environmentally sensitive locations
NP04: Reduced likelihood of environmental accidents through process improvements
NP05: Reduced waste by streamlining processes
NP06: Used waste as inputs for own processes
NP07: Disposed of waste responsibly
NP08: Handled or stored toxic waste responsibly
3. Social performance (SP)
SP01: Considered stakeholder interests in investments by creating a formal dialogue
SP02: Communicated the firm's environmental impacts and risks to the public
SP03: Improved employee or community health and safety
SP04: Protected claims and rights of local community
SP05: Showed concern for the visual aspects of the firm's facilities and operations
SP06: Recognized and acted on the need to fund local community initiatives

Appendix B. Evaluation of the measurement model

Table B1
Indicator loadings for the reflective items and constructs

Item	latent variables	loading	Item	latent variables	loading
PC101	PC1	0.83	NP02	NP	0.79
PC102	PC1	0.80	NP03	NP	0.82
PC103	PC1	0.91	NP04	NP	0.83
PC201	PC2	0.78	NP05	NP	0.86
PC202	PC2	0.81	NP06	NP	0.83
PC203	PC2	0.85	NP07	NP	0.82
PC301	PC3	0.79	NP08	NP	0.96
PC302	PC3	0.93	SP01	SP	0.83
PC303	PC3	0.88	SP02	SP	0.83
EP01	EP	0.74	SP03	SP	0.86
EP02	EP	0.81	SP04	SP	0.89
EP03	EP	0.86	SP05	SP	0.86

(continued on next page)

Table B1 (continued)

Item	latent variables	loading	Item	latent variables	loading
EP04	EP	0.83	SP06	SP	0.81
NP01	NP	0.75			

Table B2
Internal consistency reliability and convergent validity measures for the model's reflective constructs

latent variables	Cronbach's alpha	rhoC	rhoA	AVE
PC1	0.81	0.87	0.85	0.59
PC2	0.84	0.85	0.84	0.60
PC3	0.88	0.92	0.90	0.62
EP	0.83	0.89	0.88	0.62
NP	0.84	0.88	0.85	0.61
SP	0.83	0.85	0.83	0.58

AVE: Average variance explained.

Table B3
Bootstrapped heterotrait–monotrait (HTMT) values for the model's reflective constructs

Path	Original estimate	Bootstrap mean	Bootstrap SD	T	5 % CI	95 % CI
PC1→PC2	0.62	0.71	0.11	1.89	0.56	0.88
PC1→PC3	0.51	0.53	0.09	3.26	0.38	0.72
PC1→EP	0.63	0.66	0.15	2.14	0.46	0.76
PC1→NP	0.39	0.43	0.09	1.92	0.35	0.61
PC1→SP	0.46	0.53	0.08	2.11	0.36	0.66
PC2→PC3	0.28	0.31	0.08	2.97	0.21	0.43
PC2→EP	0.33	0.37	0.06	7.89	0.25	0.46
PC2→NP	0.26	0.28	0.10	6.78	0.19	0.36
PC2→SP	0.32	0.35	0.06	6.92	0.28	0.44
PC3→EP	0.12	0.16	0.09	1.92	0.08	0.29
PC3→NP	0.19	0.21	0.06	1.26	0.11	0.35
PC3→SP	0.33	0.36	0.03	2.01	0.28	0.53
EP→NP	0.32	0.35	0.13	4.32	0.26	0.41
EP→SP	0.38	0.41	0.12	3.45	0.32	0.54
NP→SP	0.28	0.31	0.12	2.21	0.24	0.36

Data availability

The data that has been used is confidential.

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